

Noisy Controls in Two-Pass Regressions

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Note: This proposal is a preliminary document and may contain inadvertent errors or omissions, which will be addressed and refined in subsequent versions.

Abstract

Fama-French factors are ubiquitous in finance, mainly acting as controls in linear settings. Notably, their usage has been tantamount in the two-pass cross-sectional regressions employed in asset pricing tests. While the issue of error in variables (EIV) in such procedures has been studied extensively, less is known about the impact of the rounding errors in Fama-French factors often acting as controls in the first pass of the regression. Specifically we demonstrate that the rounding errors induced by the decimals approximation implied in the construction of the Fama-French factors also contributes to incorrect inference by creating an EIV bias in the first pass of the regression. We further show empirically the impact of such bias on a set of clear predictors of excess returns.

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1 Introduction

Traditional methodology for anomaly detection in asset pricing usually involves one of two procedures:

1. Fama-Macbeth (1973) (FM) two-stages least square regression with the suggested anomaly portfolio time-series as independent variable and accompanying controls. If the coefficient $\hat{\beta}$ on the portfolio y is marginally significant given a confidence level (usually $p = 0.05$), it suggests that the component is priced in the excess returns of the market and hence can be considered an anomaly.

$$r_{i,t} = \alpha_{0,i} + \beta_{1,i}\text{RM}_t + \beta_{2,i}\text{SMB}_t + \beta_{3,i}\text{HML}_t + \beta_{4,i}y_t + \epsilon_{i,t} \quad (1)$$

2. Sorting the cross-section of stocks on the desired variable, creating a long-short portfolio that expresses the premia associated with this variable and testing it for significant alpha using ordinary least squares. If the intercept $\hat{\alpha}$ is significant, it suggests that the portfolio displays non-zero excess returns above the benchmark and can therefore be considered a predictor.

$$y_t = \alpha_0 + \beta_1\text{RM}_t + \beta_2\text{SMB}_t + \beta_3\text{HML}_t + \epsilon_t \quad (2)$$

where y_t represents the anomaly premia at time t and $r_{i,t}$ the return of portfolio or asset i at time t . Literature on the improvement of statistical methodology in those procedures has flourished recently. Several initiatives have been directed towards reducing omitted variable bias by introducing summarized versions of other factors as control sets in addition to the conventional FF specification. Put simply, any of the aforementioned test fails to introduce the appropriate controls in its setting without losing parsimony. As the set of controls should theoretically include all the significant predictors (and possibly others that are yet undiscovered) to accurately assess the marginal significance of the tested predictor, previous configurations de facto introduce an omitted variable bias which affects conclusions drawn from hypothesis testing. Solutions most often favored to partially solve this concern usually consist in the application of a factorization method applied to knowingly significant predictors to generate the optimal control set as benchmark without losing parsimony of the model (B. T. Kelly, Pruitt, and Su, 2019; Gu, B. Kelly, and Xiu, 2021; Giglio and Xiu, 2021).

Another important problem that has been brought to attention is the errors-in-variables (EIV) bias emerging from the traditional two-pass procedure employed in (1). Formally, solving for factor risk premium in this setting requires solving the two-pass cross sectional regression that follows (IHS EViews, 2014). The first step is done to obtain the factor exposures:

$$\begin{aligned} r_{1,t} &= \alpha_1 + \beta_{1,F_1} F_{1,t} + \beta_{1,F_2} F_{2,t} + \dots + \beta_{1,F_m} F_{m,t} + \epsilon_{1,t} \\ r_{2,t} &= \alpha_2 + \beta_{2,F_1} F_{1,t} + \beta_{2,F_2} F_{2,t} + \dots + \beta_{2,F_m} F_{m,t} + \epsilon_{2,t} \\ &\vdots \\ r_{n,t} &= \alpha_n + \beta_{n,F_1} F_{1,t} + \beta_{n,F_2} F_{2,t} + \dots + \beta_{n,F_m} F_{m,t} + \epsilon_{n,t} \end{aligned} \tag{3}$$

where $r_{i,t}$ is the return of portfolio or asset i (n total) at time t , $F_{j,t}$ is the factor j (m total) at time t , β_{i,F_m} are the factor exposures, or loadings, that describe how returns are exposed to the factors, and t goes from 1 through T . The second step is employed to compute the exposure of returns to the factor loadings:

$$\begin{aligned} r_{i,1} &= \gamma_{1,0} + \gamma_{1,1} \hat{\beta}_{i,F_1} + \gamma_{1,2} \hat{\beta}_{i,F_2} + \dots + \gamma_{1,m} \hat{\beta}_{i,F_m} + \varepsilon_{i,1} \\ r_{i,2} &= \gamma_{2,0} + \gamma_{2,1} \hat{\beta}_{i,F_1} + \gamma_{2,2} \hat{\beta}_{i,F_2} + \dots + \gamma_{2,m} \hat{\beta}_{i,F_m} + \varepsilon_{i,2} \\ &\vdots \\ r_{i,T} &= \gamma_{T,0} + \gamma_{T,1} \hat{\beta}_{i,F_1} + \gamma_{T,2} \hat{\beta}_{i,F_2} + \dots + \gamma_{T,m} \hat{\beta}_{i,F_m} + \varepsilon_{i,T} \end{aligned} \tag{4}$$

where the returns r are the same as those in (3), γ are regression coefficients that are later used to calculate the risk premium for each factor, and in each regression i goes from 1 through n . It is evident to see from these systems of equations that the explanatory variables in (4) are measured with error, as they are results of the previously obtained coefficients from (3). Shanken (1992) suggests a modified version of the traditional two-pass procedure which allows for a multifactor setting (such as the one outlined previously). Kim (1995) proposes a new correction for the EIV problem employing maximum likelihood estimation under either homoscedasticity or heteroskedasticity. Numerous other further studies explore the problem of weak identification in the context of asset-pricing two-pass regressions (Kan, Robotti, and Shanken, 2013; Gospodinov, Kan, and Robotti, 2014; Khalaf and Schaller, 2016).

Our approach slightly departs from the previously discussed research when we specifically ask the question: what if the factors employed in the first pass of the FM procedure were themselves plagued by EIV?

2 Methodology

As we know from the Fama-French Library (2023), available factors are constructed and then rounded to the second decimal point. In traditional two-pass regressions, measurement error of the beta variable is the same as the estimation error of the beta estimated in the first-pass regression, which has allowed for researchers to suggest credible corrections to the EIV problem (Kim, 1995). However, we emphasize that the error created by rounding the Fama-French factors is an unknown quantity which cannot be modeled accurately. Hence, while some EIV corrections might be N -consistent (with a fixed T and a number of assets that goes to infinity), no such correction is possible for the measurement error created by rounding the FF factors. An alternative solution might be that each set of researchers replicate the Fama-French factors to allow for full precision. However, we know from past research that achieving full precision in replication is extremely difficult, if not impossible (Chen and Zimmermann, 2021). This solution might also open the pandora’s box of countless methodologies, which we know already constitute a problem (Menkveld et al., 2022).

To determine the impact of rounding errors on the inference generated by two-pass procedures, we suggest to start the analysis with the following empirical exercise. A very preliminary notebook which outlines this procedure is available in my [Github Portfolio](#).

1. Employ the methodology outlined in (3) and (4) to estimate the risk premia associated with a panel of anomalies gathered by (Chen and Zimmermann, 2021).
2. Bootstrap an $N \times F$ matrix from a truncated normal distribution over the tick size of the Fama-French factors acting as controls and add this to the original factors from the library. This should result in multiple panels of factors which represent the factors at high precision.
3. Repeat the first step using the high-precision panels of factors and raise the differences in results obtained.

3 Expected Results

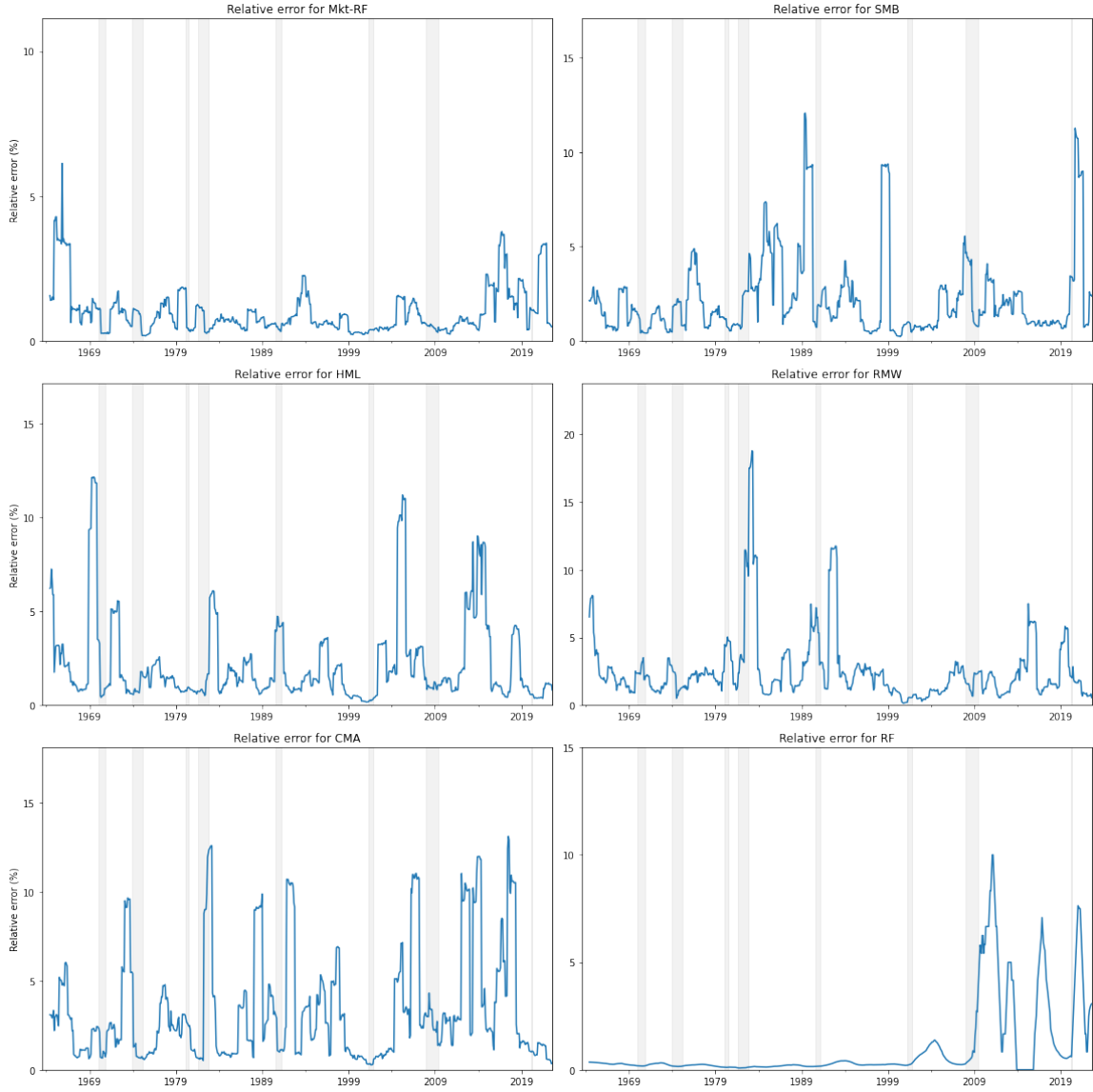


Figure 1: Relative rounding error for each component of the Fama-French 5 factors model, at the monthly frequency.

	Mkt-RF_RelativeError	SMB_RelativeError	HML_RelativeError	RMW_RelativeError	CMA_RelativeError	RF_RelativeError
N	14895	14895	14895	14895	14895	14895
Mean	0.059014	0.0829934	0.0942004	0.118209	0.115822	0.102225
Std	0.131277	0.156625	0.167915	0.189324	0.188589	0.201881
Minimum	0	0	0	0	0	0
25%	0.0107527	0.0188679	0.0208333	0.0285714	0.0277778	0.03125
Median%	0.0212766	0.0333333	0.04	0.0526316	0.0526316	0.047619
75%	0.047619	0.0714286	0.0833333	0.111111	0.111111	0.0714286
Maximum	1	1	1	1	1	1

Table 1: Specifics of relative error for each components of the FF5 factors for monthly frequency.

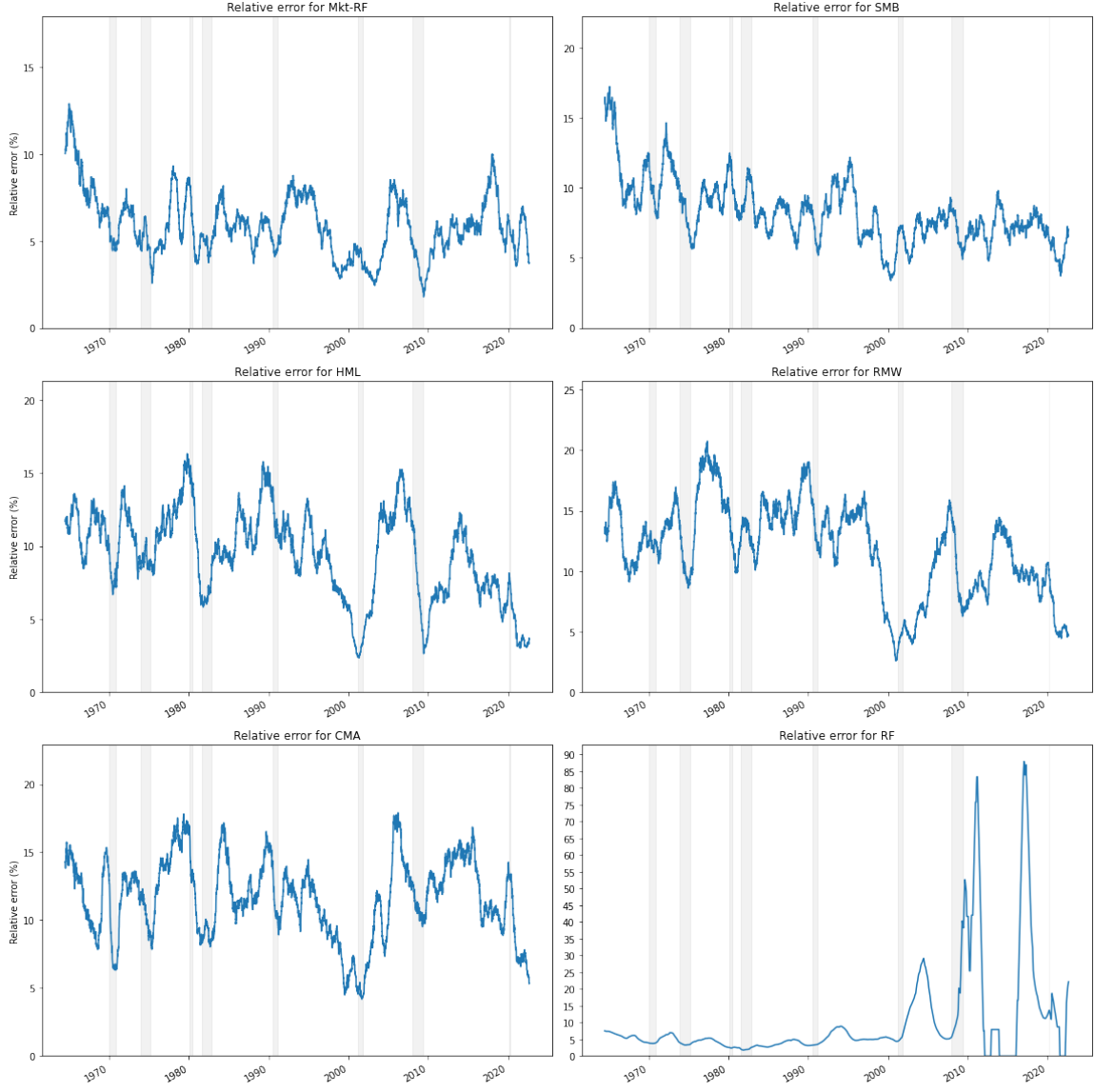


Figure 2: Relative rounding error for each component of the Fama-French 5 factors model, at the daily frequency.

	Mkt-RF_RelativeError	SMB_RelativeError	HML_RelativeError	RMW_RelativeError	CMA_RelativeError	RF_RelativeError
N	710	710	710	710	710	710
Mean	0.0100621	0.0223898	0.0221745	0.0258721	0.0333017	0.00975242
Std	0.0284018	0.0769989	0.0747897	0.0758094	0.111439	0.024029
Minimum	0	0	0.000716846	0	0	0
25%	0.00210526	0.0031008	0.0034276	0.00529101	0.00463501	0.0016129
Median%	0.00357783	0.00540541	0.00621118	0.00877193	0.00806452	0.00238095
75%	0.00724638	0.0140845	0.0138889	0.0196078	0.0172414	0.0037037
Maximum	0.333333	1	1	1	1	0.1

Table 2: Specifics of relative error for each components of the FF5 factors for daily frequency.

Based on a small notebook that is available on demand, [Figure 1](#) and [Figure 2](#) present the relative rounding error for each component of the Fama-French 5 factors model, at the monthly frequency.¹ Each relative error is computed using a tick size of 0.01, as the method of rounding employed by Fama-French is unknown. We can see that the relative error rate hovers around 2% for the monthly frequency and around 10% for the daily frequency. These quantities would therefore be what constitutes the EIV bias for the FF factors in the first pass of the two-pass regression outlined above. Descriptive statistics for each set of relative errors are also outlined.

Following these preliminary results, I believe that there is sufficient grounds to conduct a more extensive analysis of this problem.

¹Note that this is in exception to the RF, which is not a factor, but we add for completeness.

4 References

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